Policy-Constrained Generation Portfolios and Transmission Congestion Mitigation in Minas Gerais, Brazil

A Two-Horizon, Two-Scope Computational Screening Study

Working paper — authorship and affiliations to be confirmed
Not peer reviewed; not an approved engineering investment plan

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Abstract

The transition from state-level renewable-generation targets to a physically deliverable electricity system requires both investment planning and network feasibility assessment. We develop a sequential, auditable screening framework for Minas Gerais (MG), Brazil, at 2035 and 2050 horizons. Part I solves policy-constrained, 8,760-hour generation and storage capacity-expansion linear programs, then assigns fixed incremental portfolios to candidate substations by mixed-integer optimization. The resulting modeled additions are 3.217 GW in 2035 and 30.101 GW in a research-defined 2050 local fossil-free and zero-import scenario, including 5.290 GW of battery power capacity. A four-area northern siting case is compared against optimization over all 12 candidate resource areas, while retaining the same 1,489-bus, 1,890-branch network for both cases. Part II performs incremental DC/PTDF thermal screening on 283 filtered high-voltage lines and explicitly updates the network admittance matrix when illustrative parallel circuits are added. Under MG-local balancing, the regional siting alternative reduces indicative 2035 connection expenditure from R$1,357.88 million to R$878.32 million, and 2050 connection expenditure from R$5,359.38 million to R$4,607.66 million. For 2050, seven modeled parallel circuits reduce the worst screened overload aggregate to 160.69 MW (local) and 164.76 MW (regional), but do not eliminate violations. Sensitivity to an assumed external balancing bus is substantial. The exercise demonstrates the value of broader siting flexibility and the risk of treating sequential DC screens as fully secured expansion plans: the 2050 topology is not observed, residual equipment bottlenecks remain, and AC, contingency, hourly network operation and project-level cost verification are incomplete.

Keywords: capacity expansion; power transmission; mixed-integer optimization; PTDF; congestion mitigation; Minas Gerais; renewable energy; interconnection costs.

1 Background and research motivation

1.1 Why generation portfolios alone are not sufficient

Brazil's integrated power system already accommodates a substantial share of renewable electricity, while expansion of variable generation and distributed resources creates increasingly important questions about where additional capacity can be connected and how its output will be transported. The federal *Plano Decenal de Expansão de Energia 2035* (PDE 2035), prepared by Empresa de Pesquisa Energética (EPE) under the Ministry of Mines and Energy, provides an official planning context for national energy supply, demand and transmission expansion [3, 5].

Those national planning scenarios should not be confused with a feasible generation or network buildout for any particular state.

The state of Minas Gerais is a suitable case study because the *Plano Estadual de Ação Climática* (PLAC-MG) sets a 2050 net-greenhouse-gas-emissions objective and contains technology-specific renewable-power actions, notably action 7.3.2.4 [1, 2]. State-wide net-zero emissions, however, do *not* logically require zero fossil generation in every operating hour; imports, residual thermal generation, bioenergy life-cycle emissions and carbon accounting must be considered separately. We therefore label the 2050 zero-local-gas/zero-import case in this study an explicit *research counterfactual*, not a binding statutory power-sector mandate.

A conventional aggregate least-cost generation portfolio may satisfy annual policy milestones and chronological supply–demand constraints yet overload a particular line after projects are assigned to buses. In turn, minimizing the straight-line distance from a resource region to a substation can be misleading when the resulting injections alter regional transfer flows. Such interactions motivate the combination of hourly energy-system optimization, nodal siting, DC/PTDF screening and explicit network reinforcement. DC power flow is widely employed for such studies but may misrepresent AC voltages, reactive power, transformer capability and contingency risk [7, 8]. The larger transmission expansion literature also distinguishes globally optimized mixed-integer investment models from heuristic candidate selection [9].

1.2 Research questions and contribution

The study addresses three questions: (i) What policy-constrained incremental generation/storage portfolios arise in the adopted 2035 and fossil-free 2050 scenarios? (ii) How much does the ability to choose interconnection sites across the existing candidate regions reduce indicative connection expenditure relative to a restricted four-area case? (iii) What screened overloads remain after hypothetical parallel-line construction and how sensitive are the results to the treatment of power balancing?

Our contribution is primarily *methodological and diagnostic*, rather than an engineering-approved plan. We (1) document fixed two-horizon hourly portfolio solutions, (2) compare local and regional siting freedom against exactly the same full regional network, and (3) track how an actual change to DC network topology affects congestion. The numerical work is explicitly separated from additional data collection required for construction-grade decision-making.

1.3 Study boundary and terminology

A *full-region siting case* here means free siting across the study's *12 preselected resource areas*, not unconstrained access to every possible MG land parcel or substation. A *monitored-line overload* is an active-power, DC-screen exceedance using the available MVA rating as a proxy for a MW limit. A *thermal-uplift variable* is a soft continuous relaxation in the siting optimization, not a physically specified upgrade. Finally, *connection cost* is a simplified estimate for linking candidate resources to qualifying stations and is different from the present-valued macro generation-investment objective.

2 Data, modeling boundaries, and computational design

2.1 Sources and data hierarchy

Table 1 gives the operational role of the main inputs. Historical and official sources are separated from analytic choices. The electricity planning data are a single reference network rather than

an ensemble of realized 2035/2050 operations. The 2026 substation-coordinate crosswalk is not an authorization to build a facility there.

Table 1: Inputs and the distinct inferential roles assigned to them.

Input family	Use in this study	Crucial qualification
PLAC-MG / MRV [1, 2]	Technology-specific tracking milestones; 2035 interpolated research target	No standalone observed 2035 milestone
ONS historical time series	Hourly demand, resource shapes and observed hydro envelopes	Historical 2025 patterns are not 2050 weather forecasts
EPE PDE 2035 / PWF [3, 4]	1,489 buses; 1,890 branches; 2035 reference flows and parameters	No native validated 2050 case
ONS coordinates / study crosswalk	Candidate region–substation distances; 101 mapped MG buses	Site eligibility and connection capacity unverified
EPE transmission/cost studies [5, 6]	Selected 230/500-kV unit-cost references	138/345-kV and associated equipment have model assumptions
Study-defined 12 candidate areas	Siting test set; four northern areas define restricted case	Not verified plant projects or regional potential limits

The reference EPE network contains 1,489 buses and 1,890 branches. Of these, 283 filtered high-voltage physical lines with usable thermal attributes are screened. This selection excludes many devices, lower-voltage elements and incompletely mapped records: passing all 283 checks does *not* imply full network feasibility. The 2050 tests retain the 2035 PWF topology as a *frozen-network stress experiment.*

2.2 Optimization sequence and scenario definitions

Figure 1 displays the methodological distinction between the two parts. First, an aggregate chronological LP defines installed technology quantities. Next, a nodal mixed-integer program assigns these *fixed* quantities to candidate locations. Finally, explicit parallel circuits are tested by re-solving DC flows. None of the last-stage congestion benefits feeds back into a globally joint capacity, location and line-investment program in this draft.

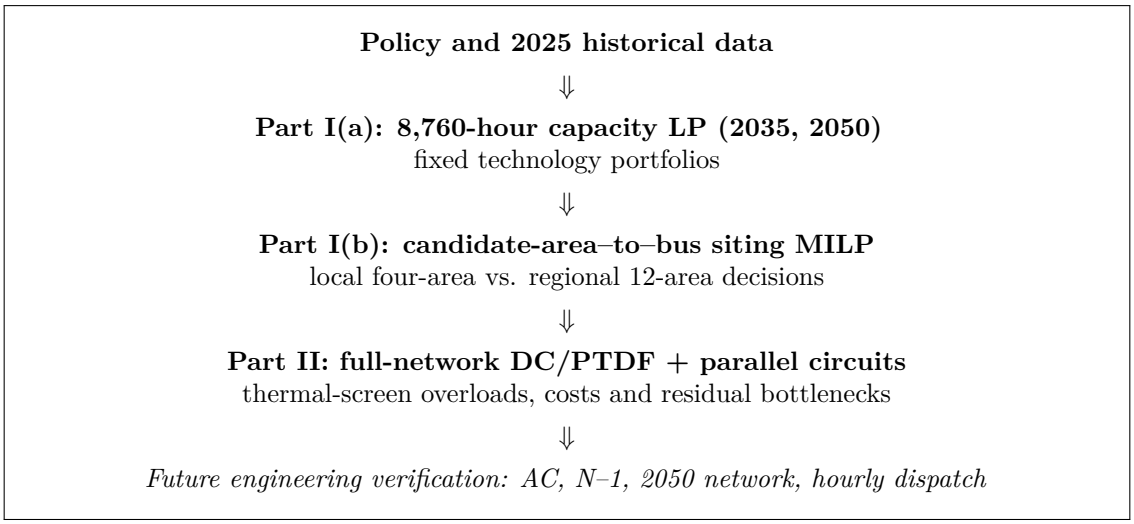


Figure 1: Sequential computational study. The final engineering verification box is a research requirement, not a completed stage.

For both years, photovoltaic coincidence factors of 35%, 70% and 90% generate three local-load-balancing cases. A six-case stress set additionally allows incremental power to be withdrawn

at external bus 585. The latter is a transmission sensitivity, *not* a statement that 2050 will export every incremental MW nor a requirement implied by the 2050 no-import assumption.

3 Part I: policy-constrained portfolios and interconnection siting

3.1 Chronological capacity-expansion linear program

Let $k \in \mathcal{K}$ index technologies, $t \in \{1, \ldots, 8760\}$ index representative annual hours and $y \in \{2035, 2050\}$ a *separately solved* terminal horizon. The first stage chooses new capacity $K_{k,y} \geq 0$, hourly renewable use v_t, hydro h_t, gas g_t, biogas b_t, imports m_t, storage charge c_t, discharge d_t, state of charge E_t and unserved energy q_t. Its research objective is an incremental discounted technical-cost proxy,

$$\min \sum_{k \in \mathcal{K}} C_{k,y}^{\text{cap+fixed}} K_{k,y} + \sum_t w_y \big(c_{\text{imp}} m_t + c_{\text{gas}} g_t + c_{\text{bio}} b_t + \epsilon (c_t + d_t) \big), \tag{1}$$

where the investment coefficients include asset life, fixed O&M, a discounting convention and an estimated average interconnection adder from an earlier siting iteration. The electricity-balance equation is

$$h_t + v_t + g_t + b_t + m_t + d_t + q_t = L_{y,t} + c_t. \tag{2}$$

The pooled nondispatchable term v_t includes an observed existing supply template plus the renewable profiles of surviving additions; existing distributed PV is included only once. Representative resource bounds include

$$0 \leq v_t \leq v_t^{\text{existing}} + \sum_{k \in \mathcal{K}_{\text{VRE}}} \phi_{k,t} K_{k,y}, \qquad 0 \leq b_t \leq \bar{\phi}_{\text{bio}} K_{\text{biogas},y}. \tag{3}$$

Hydro uses historical hourly upper envelopes, calendar-month energy budgets and an empirical ramp constraint; no reservoir inflow state is inferred. Storage obeys a single-converter and energy-state formulation:

$$\begin{aligned} E_t &= E_{t-1} + \sqrt{\eta_{\text{rt}}}\, c_t - \frac{d_t}{\sqrt{\eta_{\text{rt}}}}, \\ 0 \leq E_t \leq D K_{\text{BESS},y}, &\qquad c_t + d_t \leq K_{\text{BESS},y}, \end{aligned} \tag{4}$$

with a cyclic annual boundary condition. The reported 2050 scenario uses $D = 8$ hours. A modeling floor $K_{k,y}^{\text{existing}} + K_{k,y} \geq P_{k,y}$ is imposed on the four PLAC-tracked categories (central PV, distributed PV, biomass, distributed/small hydro). For 2035, $P_{k,2035}$ is an interpolation between published 2030 and 2050 tracking endpoints, rather than a separately published legal target.

In the 2050 counterfactual, explicit local gas dispatch and external imports are constrained to zero, while dispatchable biogas is allowed. The model does not represent the full life-cycle emissions of biomass or grid purchases and does not certify an economy-wide climate objective. The modeled optimal shortage variable q_t is zero in the selected historical-template runs; this is *not* a probabilistic reliability result or evidence of N–1 security.

3.2 Fixed optimized portfolios at the two horizons

The resulting incremental capacities, used as *fixed inputs* to all spatial comparisons below, are reported in Table 2 and Fig. 2. Values for central and distributed solar refer to the new-build portion, not the entire existing installed fleet. Storage is represented in GW power and separately in GWh energy.

Table 2: Frozen technology additions used in the nodal and transmission study; GW unless noted.

Technology	2035 addition	2050 addition
Centralized solar PV	1.000	12.918
Distributed solar PV	1.044	7.944
Biomass	0.877	0.877
Small hydro	0.296	0.671
Candidate-site wind	0.000	2.000
Biogas	0.000	0.400
Battery storage (GW)	0.000	5.290
Total incremental GW	**3.217**	**30.101**
Battery energy (GWh)	0.000	42.324

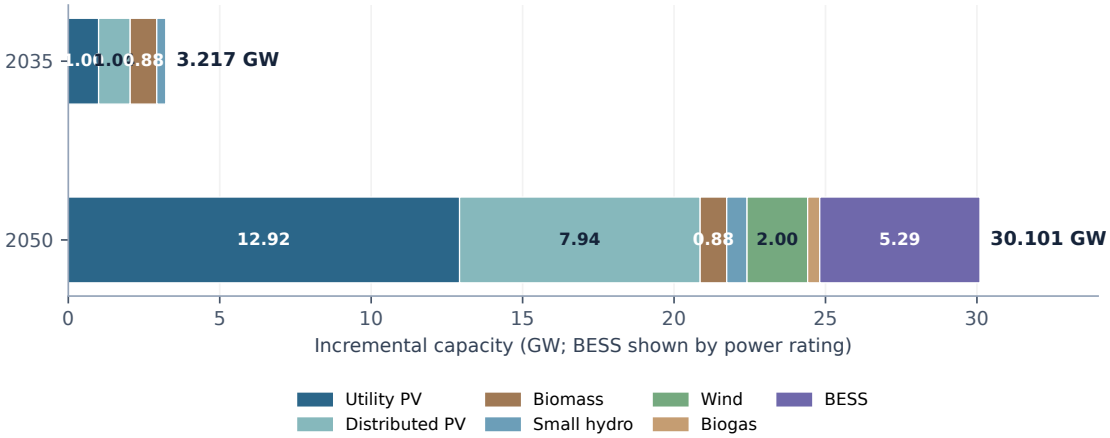


Figure 2: Fixed incremental capacities by technology. Battery storage is expressed in power units (GW), not energy (GWh). The 2050 wind and biogas resources are capped research candidates, not commissioned projects.

The 2035 LP represents approximately 80.425 TWh annual modeled load and 2.293 TWh of modeled imports. Its objective is approximately R$15.753 billion in end-2025 present value using constant-2024-BRL input conventions. The 2050 run represents 97.186 TWh of modeled load, zero modeled imports and zero explicit local gas dispatch, 15.012 TWh of potential renewable curtailment and 5.290 GW / 42.324 GWh of new batteries; its analogous objective is approximately R$20.285 billion. Both objectives use a specified model boundary and should not be interpreted as full state-wide electricity-system investments. In particular, they exclude complete renewal of the existing fleet and verified construction-level network costs.

3.3 Decision scopes and candidate substations

The second stage keeps $K_{k,y}$ fixed while choosing the allocation x_{as} (MW) from resource cluster a to eligible grid station s, along with an integer number z_{as} of interconnection circuits. Each resource area has at most three geographically and electrically matched candidate stations in the input set. Across the 12 areas the crosswalk contains 101 mapped MG buses, but only subsets are eligible for each cluster.

In the **restricted local case (A)**, the four northern candidate resource areas Arinos, Janaúba, Jaíba and Montes Claros can change allocation and connection topology; all other clusters are held at their nearest eligible baseline station. In the **regional candidate case (B)**, all modeled clusters may be allocated across their eligible stations. Both cases are evaluated on the same complete 1,489-bus network. In 2035, only ten candidate areas have nonzero supported development in the current portfolio; in 2050, all twelve are represented.

3.4 Siting and thermal-relaxation MILP

Define c_{as}^{var} as estimated MW-proportional connection expense, c_{as}^{circ} as a fixed circuit/bay contribution and $u_\ell \geq 0$ as a *continuous* thermal-rating slack for monitored line ℓ. The screened siting problem is

$$
\begin{aligned}
\min_{x,z,u} \quad & \sum_{(a,s)} c_{as}^{\mathrm{var}} x_{as} + \sum_{(a,s)} c_{as}^{\mathrm{circ}} z_{as} + \lambda \sum_{\ell \in \mathcal{L}_{\mathrm{mon}}} u_\ell, \\
\text{s.t.} \quad & \sum_{(a,s):k(a)=k} x_{as} = 1000 K_{k,y}^{\star}, \\
& \sum_{s \in \mathcal{S}_a} x_{as} \leq 1.5\, \bar{K}_a, \quad 0 \leq x_{as} \leq \bar{P}_{as} z_{as}, \quad z_{as} \in \mathbb{Z}_{\geq 0}, \\
& -\bar{F}_\ell - u_\ell \leq F_{\ell,\omega}^{0} + \sum_{(a,s)} H_{\ell as,\omega} x_{as} \leq \bar{F}_\ell + u_\ell \quad \forall \ell, \omega.
\end{aligned}
\tag{5}
$$

The $1.5\,\bar{K}_a$ cap is an assumed upper bound relative to reference geographic allocation, not verified developable resource. The screening penalty $\lambda = 1$ million R$/MW is a tuning parameter to discourage overload; it is explicitly *not* a construction-cost quotation. The matrix H is built from 2035 DC PTDFs under the corresponding balancing convention. Distributed PV enters as fixed spatial injections in the siting screen where appropriate; battery net injection is set to zero rather than assuming arbitrarily favorable single-hour battery dispatch.

Connection distances use coordinates for candidate area centers and candidate substations. Route length is approximated by $d_{as}^{\mathrm{route}} = 1.25\, d_{as}^{\mathrm{straight}}$, multiplied by line unit rates and relevant bay costs. The source cost notebook uses EPE-derived 230/500-kV line references and explicitly assumed proxy values for 138/345 kV, connection capacity, transformers and distribution reinforcement [6]. Interconnection costs are 2025-R$ *undiscounted indicative* estimates, not directly additive to the macro LP's differently timed present value.

3.5 Siting optimization results

Table 3 summarizes the base MG-local balancing results. All eight siting MILPs (two years, two scopes, two scenario sets) returned an optimal solver termination; the saved maximum linear-constraint residual was below 4×10^{-9} in the solver units. This is numerical feasibility of the specific MILP, not validation of site permits or thermal ratings.

Table 3: Part I siting results under MG-local balancing. Connection costs are indicative, undiscounted million R$; u is an artificial aggregate thermal-uplift proxy (MW).

Horizon	Decision scope	Connection cost	$\sum_\ell u_\ell$
2035	A: four northern areas	1,357.88	0.00
2035	B: all eligible areas	**878.32**	0.00
2050	A: four northern areas	5,359.38	**663.31**
2050	B: all eligible areas	**4,607.66**	677.96

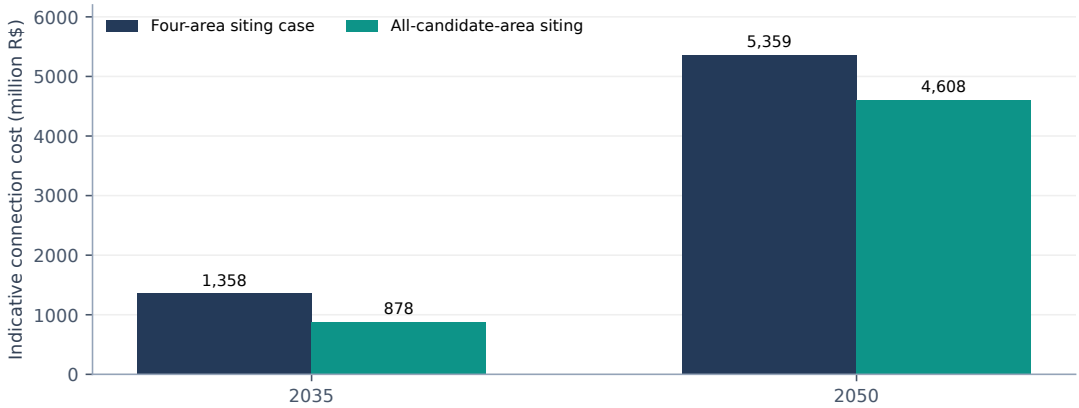


Figure 3: Siting cost of fixed portfolios for the local and regional alternatives, excluding generation CAPEX and physical network reinforcement. Broader siting freedom reduces modeled 2035 and 2050 connection costs by 35.3% and 14.0%, respectively, relative to the restricted case.

The regional decision set reduces 2035 connection costs by R$479.56 million (35.3%) relative to the restricted northern alternative. For 2050 the corresponding reduction is R$751.73 million (14.0%). Since the 2050 objective jointly trades connection cost against a soft overload penalty, the cheaper regional solution has a slightly *larger* thermal-uplift proxy: 677.96 MW versus 663.31 MW. That tradeoff is economically consistent with the model and does not make the regional case a pointwise minimum of every individual risk metric.

4 Part II: transmission congestion screening and relief

4.1 Full-network DC/PTDF diagnostic

The available EPE planning network is represented by buses $\mathcal{N}$, existing branches $\mathcal{E}$ and bus–branch incidence matrix A. With reactance X_ℓ and $b_\ell = 1/X_\ell$, the DC bus matrix is $B_{\text{bus}} = A^\top \text{diag}(b) A$, with a reference angle removed. The base-calibrated incremental flow sensitivity satisfies

$$\Delta f = \text{PTDF}\, \Delta p, \qquad f_{\ell\omega} = f_\ell^{\text{ref}} + \Delta f_{\ell\omega}, \qquad \mathcal{O}_\omega = \sum_{\ell \in \mathcal{L}_{\text{mon}}} \left[|f_{\ell\omega}| - \bar{F}_\ell \right]_+ . \tag{6}$$

The screen checks the *worst-case sum across lines within one scenario*, $\max_\omega \mathcal{O}_\omega$. It is not a sum of simultaneously occurring overloads across all scenarios, annual energy not served or yearly

curtailed energy. The constrained set has 283 line records out of 1,890 total branches. The ratings are used as MW screening proxies even where originally specified in MVA; AC validation is thus essential.

4.2 Explicit parallel-circuit experiment

The Part I u_ℓ slack does not alter reactance or network flows and cannot represent real construction. Part II therefore selects existing line corridors with both endpoints geographically mapped and evaluates an additional parallel circuit with the corridor's reference reactance and thermal rating. Each accepted circuit contributes an extra branch to the incidence matrix, and the bus susceptance matrix is rebuilt. The DC nodal balance and line flows are recomputed for *each scenario* before considering the next circuit. This procedure distinguishes a physical-topology experiment from simply increasing $\bar{F}_\ell$ in Eq. (5).

Candidate circuits are greedily ranked by reduction in the maximum scenario's screened overload sum per estimated construction cost,

$$\ell^* = \arg\max_{\ell \in \mathcal{C}} \frac{\max_\omega \mathcal{O}_\omega^{\text{current}} - \max_\omega \mathcal{O}_\omega^{\text{trial}}(\ell)}{\widehat{C}_\ell^{\text{circuit}}}. \tag{7}$$

For eligible geocoded lines, estimated construction cost is $\widehat{C}_\ell \approx c_{\text{line}}(1.25 d_{\text{direct}}) + 2c_{\text{bay}}$, with station-equipment differences simplified. The greedy step is a *heuristic* and has no joint expansion-optimality certificate. A short remaining bottleneck or a missing geocoded endpoint can fall outside the costed candidate set even when its loading is severe.

4.3 2035 versus 2050 and local versus regional siting

For 2035, neither siting alternative creates additional thermal exceedance on the monitored set under the modeled DC scenarios, and the parallel-circuit heuristic selects no new lines. This conclusion is narrow: it does not reverse the prior independent AC warnings about voltage and equipment constraints.

The results are different in 2050 (Table 4 and Fig. 4). Under local-load balancing, seven parallel circuits with a combined indicative construction estimate of R$236.93 million reduce the worst-case overload sum from 663.31 to 160.69 MW for the four-area case and from 677.96 to 164.76 MW for the regional case. On a consistent *connection-plus-circuits* screening basis, the regional estimate is R$4,844.58 million versus R$5,596.31 million, a 13.4% reduction. Neither design satisfies all monitored line limits.

Table 4: Part II 2050 results. All currency in indicative, undiscounted million R$; overload values in worst-scenario MW summed over monitored lines.

Balancing	Siting	Prebuild	Circuits	Circuit cost	Postbuild	Connection + circuits
MG load	Local A	663.31	7	236.93	160.69	5,596.31
MG load	Regional B	677.96	7	236.93	164.76	**4,844.58**
Export stress	Local A	8,779.48	12	4,424.92	170.85	9,762.09
Export stress	Regional B	8,779.48	12	4,424.92	172.31	**9,017.77**

4.4 Location of modeled bottlenecks

Under regional siting and MG-local balancing, the illustrative seven-circuit expansion leaves the largest exceedances concentrated in two existing branches at 90% PV coincidence (Fig. 5): 1646–4475 near Timóteo–Aperam, at 135.61 MW above its screening limit, and 1609–1613 near

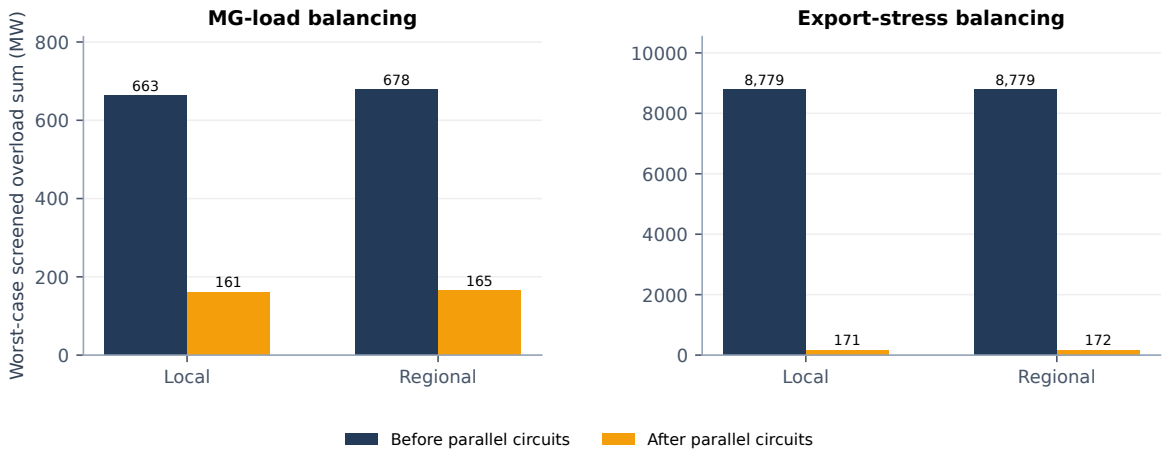


Figure 4: Screened 2050 overload before and after adding explicit modeled parallel branches. The stress-test scale differs greatly from MG-local balancing because of the assumed location of offsetting power withdrawals. Neither set is fully feasible after construction.

João Monlevade, at 29.15 MW. The restricted local siting case retains 135.61 MW and 24.45 MW on these respective links, plus approximately 0.63 MW on 39813–1525. These branches are a precise *diagnostic localization*, not a verified engineering proposal; they may represent short station connections, equipment limitations or special equivalent modeling elements.

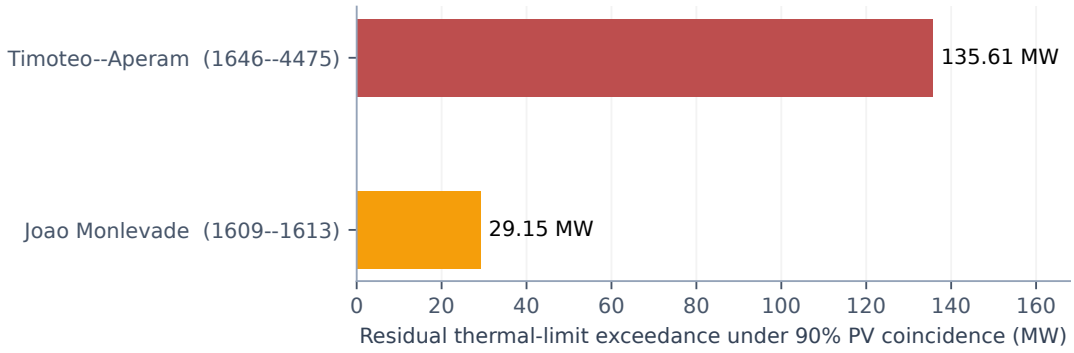


Figure 5: Regional 2050 siting case: residual line-limit exceedance after seven modeled parallel circuits under the most stressful MG-local scenario. The figures do not establish feasible physical retrofit capacities.

Table 5 lists the exact physical topology experiments under MG-load balancing in the regional case. A repeated 1623–1630 pair refers to different existing circuit reference records; a candidate branch inherits the specific record's reactance and screening rating. The estimated total should not be treated as an EPC bid, approved project or permissioned corridor.

4.5 Sensitivity to external power balancing

Introducing a single external balancing bus at 585 produces much larger modeled transfers in 2050. The greatest preconstruction overload sum rises to approximately 8,779.48 MW. The greedy model consequently tests twelve parallel circuits costing R$4,424.92 million, yet still leaves approximately 170.85 MW (local siting) or 172.31 MW (regional siting) of worst-case screened violations. Much of this additional assumed construction cost derives from long 500-kV circuits crossing the study boundary; it cannot be presented as necessary investment for a strictly zero-import 2050 case. In particular, the six-case stress set is an *injection/withdrawal sensitivity*, not a probabilistically weighted operating forecast.

Table 5: Seven illustrative parallel circuits selected by the 2050 regional MG-local heuristic.

Order	From bus	To bus	Reference circuit	Route (km)	Cost (million R$)
1	1646	26460	1	4.30	25.75
2	1623	1630	1	3.29	24.64
3	1626	26513	3	17.61	40.45
4	1620	26513	1	6.37	28.04
5	1623	26460	1	14.67	37.20
6	1626	1609	1	31.89	56.21
7	1623	1630	3	3.29	24.64
				Total	**236.93**

A separate single-hour corrective-redispatch linear screen indicates that fully removing the hardest MG-local residual thermal violations would require order-of-magnitude 8.05 GW of substitute dispatch at allowed locations under its assumptions. No physically available generator headroom, storage state of charge, water constraints or feasible multi-hour redispatch sequence was established for that test. We therefore do *not* count this optimistic dispatch bound as a completed congestion remedy.

5 Discussion, limitations, and engineering interpretation

5.1 What the comparison establishes

The restricted local problem is embedded in the regional candidate decision space: the region-wide optimization can recover the restricted allocation, or choose alternative stations when they improve the penalized objective. The observed cost saving is therefore interpretable as the value of greater *decision flexibility under this candidate set*, not a universal estimate of the value of transmission access in Minas Gerais. The residual congestion difference of roughly 4 MW between the two 2050 MG-local cases should be interpreted alongside estimated connection savings, rather than reduced to one performance ranking.

There is also an important sequencing effect. Part I selects bus injections while the PTDF matrix is fixed; Part II modifies the topology with a heuristic after the nodal decisions have been made. The resulting combined solution is *not* a globally optimal mixed-integer generation–siting–line-expansion solution. More rigorous joint planning could use explicit candidate circuits in an expanded MILP or iterative Benders-style coordination between capacity, siting and network decisions [9]. That direction requires independently substantiated circuit candidates and operating inputs.

5.2 Validity threats and missing engineering inputs

At least six limitations constrain external interpretation:

1. **Chronological versus network time resolution.** The portfolio stage is hourly, but the network phase checks selected PV coincidences around a 2035 reference state; no coupled 8,760-hour nodal dispatch or battery SOC is verified.

2. **Frozen 2050 topology.** The study does not contain an official, independently verified 2050 network case or an adopted interregional exchange schedule.

3. **DC approximation and limits.** MVA ratings are used as MW screening proxies; voltage, reactive power, transformer thermal capability, protection and N–1 security are omitted.

4. **Eligibility and physical resources.** Candidate coordinates, the 150% area ceilings, candidate-site wind/biogas potential and station connection feasibility are research assumptions, not permitted or developed projects.

5. **Incomplete project costing.** Direct distances multiplied by 1.25, unit line costs and proxy bay/transformer prices exclude terrain, right-of-way, environmental licensing, specialized station works and annual losses. Costs in Parts I and II must be aligned to a common investment base year before inclusion in an integrated objective.

6. **Carbon and reliability claims.** Zero explicit local gas and imports in the modeled 2050 run do not establish complete zero emissions; no probability of outage, Monte Carlo adequacy analysis or engineering supply-security certificate has been produced.

The next engineering data priority is to verify single-line diagrams, actual thermal ratings and retrofit choices for 1646–4475 and 1609–1613, together with an approved candidate line list, geometry, impedance, bay and transformer configurations and planning-year demand. Those inputs would permit explicit construction-choice optimization followed by AC and N–1 feasibility validation.

6 Conclusion

The completed computational study demonstrates a coherent two-horizon planning workflow, from policy-informed chronological generation portfolio to candidate-node assignment and topology-updating transmission screening. A 2035 incremental portfolio of 3.217 GW passes the selected 283-line DC thermal screen under the tested conditions. A research-defined 2050 portfolio adds 30.101 GW, including 5.290 GW of batteries, but creates substantial frozen-network congestion. Compared with a four-northern-area case, optimization across all twelve available candidate areas lowers indicative MG-local connection costs by 35.3% in 2035 and 14.0% in 2050. Seven explicit modeled parallel circuits substantially reduce 2050 MG-local overloads but fail to eliminate two dominant bottlenecks. This is a reproducible and useful planning *screen*, not a construction-ready optimum. The 2050 bus-pair bottlenecks and the strong sensitivity to the balancing boundary identify precisely where a defensible next stage of AC-validated transmission investment planning should concentrate.

Reproducibility and data statement

The calculations reported here were extracted from the project archives generated on 10–11 November 2026, which contain the two optimized hourly portfolios, 2035 reference PWF exports, candidate-area/site crosswalks, siting MILP code and outputs, topology-updating parallel-line screening code and per-scenario audit tables. The comparative siting runs terminate optimally under HiGHS, with residuals at numerical precision; the DC nodal-equation checks of the parallel-line screens are also near numerical precision. The accompanying `README.md` identifies the appropriate source archives and fixed-case results. Availability of these files does not imply that underlying EPE/ONS data can be republished under an open-data license without checking source-specific terms.

Declaration of status

This is an **internal research preprint** prepared from existing model outputs. It has not been submitted to arXiv, peer reviewed, approved by EPE/ONS, or certified for physical construction.

Author names, institutional affiliations and source-data redistribution permissions must be confirmed before public release.

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A Scenario matrix and auditing definitions

The following eight configurations constitute the principal comparison: each of two years (2035, 2050), two decision scopes (local four-area, regional candidate-area), and two scenario families (MG load only, MG load plus bus-585 export stress) is solved as a separate siting MILP and then passed through the explicit parallel-line screen. The three PV coincidence levels 0.35, 0.70 and 0.90 are evaluated in every family.

The quantity plotted as pre/post overload throughout the paper is

$$\mathcal{O}^{\mathrm{worst}} = \max_{\omega \in \Omega} \sum_{\ell \in \mathcal{L}_{\mathrm{mon}}} \max\{0, |f_{\ell,\omega}| - \bar{F}_{\ell}\}. \tag{8}$$

It does not equal the sum across *all* ω, and a reduction is not an estimate of annual curtailment. The postconstruction screen includes new circuit impedances and solves a new network model; candidate site locations and the original model's reference injection conventions remain fixed within each run.

Table 6: Full quantitative comparison of indicative connection plus modeled parallel circuit costs, million R$, and worst-case residual overload (MW).

Case / year	Scenario family	Total estimated cost	Added circuits	Residual MW
A / 2035	MG-local	1,357.88	0	0.00
B / 2035	MG-local	878.32	0	0.00
A / 2035	Export stress	1,357.88	0	0.00
B / 2035	Export stress	878.32	0	0.00
A / 2050	MG-local	5,596.31	7	160.69
B / 2050	MG-local	4,844.58	7	164.76
A / 2050	Export stress	9,762.09	12	170.85
B / 2050	Export stress	9,017.77	12	172.31